

Week 5 Report — Additional Feature Implementation

Programme: IIMSTC – IoT & Embedded Systems (Cohort 4)

Team Size: 15 Members

Hardware Team	App Development Team	Software Team
Harsha Vardhan S	Krithi	Vijetha
Ananya Chandrashekar	Nagaraju M	S Vidyashree
Palaparthi Krishna Vamsi	Charan B S	Sachidananda M M
Durga Prasad S	—	—
Ramitha R	—	—
C Pavitra	—	—
Hadiya Khan	—	—
Abhinav P	—	—

Day 1 — Sensor Logic & LCD Integration

The LCD display was successfully integrated into the system to provide real-time visualization of sensor data and system status. Code logic was developed for continuous temperature monitoring using the DHT22 sensor, ensuring consistent and reliable data acquisition.

Day 2 — Peltier Module & GSM Logic Development

The implementation of the Peltier module was completed and tested to ensure accurate thermal response. Logic for the GSM module was developed to enable communication and trigger alerts during temperature threshold violations. App development also progressed with enhancements to the alert notification interface, focusing on clarity and responsiveness.

Day 3 — Relay Module & Interface Development

The relay module was integrated into the system to enable controlled switching of connected components based on temperature conditions. Parallel development efforts focused on designing the website user interface and implementing the corresponding frontend structure. Improvements

were made to the app dashboard layout, enhancing user interaction and ensuring a more intuitive presentation of system data.

Day 4 — Buzzer Alert & Backend Setup

A buzzer alert mechanism was implemented to provide immediate audible notification when temperature levels exceed defined limits. Backend development was initiated with the creation of API keys and configuration of Firebase services, laying the groundwork for cloud-based data handling and integration.

Day 5 — Feature Integration Overview

Week 5 focused on enhancing system functionality through the integration of additional features across hardware, software, and application layers. The system evolved with improved monitoring, alert mechanisms, and user interface capabilities.

Day 6 — Review & System Analysis

All work completed during the week was reviewed and analyzed. The team evaluated system performance, integration flow, and feature functionality to ensure readiness for further development and full system integration.

Summary

Week 5 emphasized feature expansion and system enhancement. Key modules including sensors, control systems, alert mechanisms, and interface components were successfully integrated, strengthening the overall functionality and moving the system closer to a fully operational solution.